Turkey Branch Watershed Digital Twin

Model description, data, calibration and uncertainty

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Abstract

The Turkey Branch digital twin forecasts streamflow, stage, soil moisture and groundwater in the Turkey Branch watershed (Montgomery County, Maryland; 7.1 km^2 above USGS gage 01647850) for the next seven days, every six hours, with 95 % parameter and prediction intervals. It runs a physically based, semi-distributed model built in the open-source OpenHydroQual (OHQ) framework (Massoudieh et al., 2023) with 19 sub-catchments of 0.15 km^2 to 0.76 km^2. Each sub-catchment represents its connected impervious and pervious land, three soil layers and an unconfined aquifer; eight drain to open channels whose sections are measured from 1 m lidar, eleven through streets and storm sewers sized from the county and state storm-drain inventories, and three stormwater ponds are represented. Ten watershed-wide parameters were estimated by Bayesian inference (Markov chain Monte Carlo) from four years of discharge and two years of stage at the gage; the twin runs the most probable parameter set and 200 sets drawn from the posterior distribution, whose spread gives the parameter interval; the calibrated error model adds the prediction interval, which contains 93–94 % of the measurements. Tested on four further years that were not used in calibration, the model reproduces the logarithm of flow with a Nash–Sutcliffe efficiency of 0.83 and stage with 0.79. This document describes the model, its data, its calibration and the validation results.

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1 What the twin does

Every six hours the twin

1. updates its rainfall with NOAA's Multi-Radar/Multi-Sensor (MRMS) radar–gauge precipitation analysis (hourly, 1 km, about one hour behind real time) and its reference evapotranspiration with gridMET, followed by the most recent days of Open-Meteo;
2. advances the model from its last saved state to the present (the *analysis*);
3. runs the model seven days ahead from that state with the Open-Meteo hourly precipitation and reference-evapotranspiration forecasts (the *forecast*);
4. repeats both steps for 200 parameter sets drawn from the calibration's posterior distribution, each from its own saved state, and takes the 2.5 and 97.5 percentiles of their results at every time (the *95 % band*);
5. publishes the results: maps of each model element, time series of every variable in every sub-catchment with its band, and model versus USGS real-time flow and stage at the gage.

The model state is carried from cycle to cycle, so the twin keeps a continuous memory of soil moisture and groundwater. It does not assimilate observations: the model is not adjusted to match the gage, which makes the comparison with the gage shown in the viewer an ongoing, independent check.

The two 95 % intervals. The line shown for every variable is the model with the most probable parameter set (Table 2). The viewer can shade two 95 % intervals around it (box "95 % intervals", unchecked by default):

- **Parameter interval** (darker), for every variable: the 2.5 and 97.5 percentiles of the 200 runs with posterior parameter sets (Section 6.2). It shows how uncertain the output is because the parameters are only known within a range. It is narrow, about ±4 % of the flow, because the gage data determine the parameters well.
- **Prediction interval** (lighter), for streamflow in every reach and for the stage at the gage: the parameter interval combined with the error that remains between the model and the measurements, estimated in the calibration as a multiplicative, log-normal error with a standard deviation of 0.54 for the logarithm of flow and 0.17 for the logarithm of stage. The interval is the model value times $\exp(\pm 1.96\,\sigma)$, with $\sigma^2 = \sigma_e^2 + \sigma_p^2$ the sum of the error variance σ_e^2 and the variance σ_p^2 implied by the parameter interval: about a factor of 2.9 either side of the flow and of 1.4 either side of the stage. It represents everything the model does not resolve — model structure, rainfall and gage measurement — and is the range in which the gage reading is expected 95 % of the time: 93–94 % of the measured values fall inside it, in the calibration as well as in the validation years (Section 7.1). For reaches away from the gage the error of the gage is assumed, which cannot be checked there. In the forecast the interval does not include the additional error of forecast rainfall, so it understates the uncertainty of the forecast.

Flow is shown on a logarithmic axis so that both intervals can be read at low flow as well as in storms. No error model exists for soil moisture, groundwater and the other unobserved quantities, so they show the parameter interval only.

2 The watershed

Turkey Branch is a tributary of Rock Creek in the Maryland Piedmont (Aspen Hill and Rockville area). Above the USGS gage near Rockville (01647850) it drains 7.1 km^2. The basin is suburban: 39 % of it is impervious (roads 8.5 %, buildings 11.8 %, other paved areas 14.4 %, tree canopy over pavement 4.8 %), 38 % turf and 17 % forest. Only 6.4 km of stream is mapped in it; much of the drainage runs through 55 km of storm pipes. Montgomery County restored the main stem along the Matthew Henson Trail in about 2006–2009 and has since built or retrofitted several stormwater ponds; the USGS record starts in November 2006, so the calibration and validation years describe the restored channel. There are no combined sewers or reservoirs.

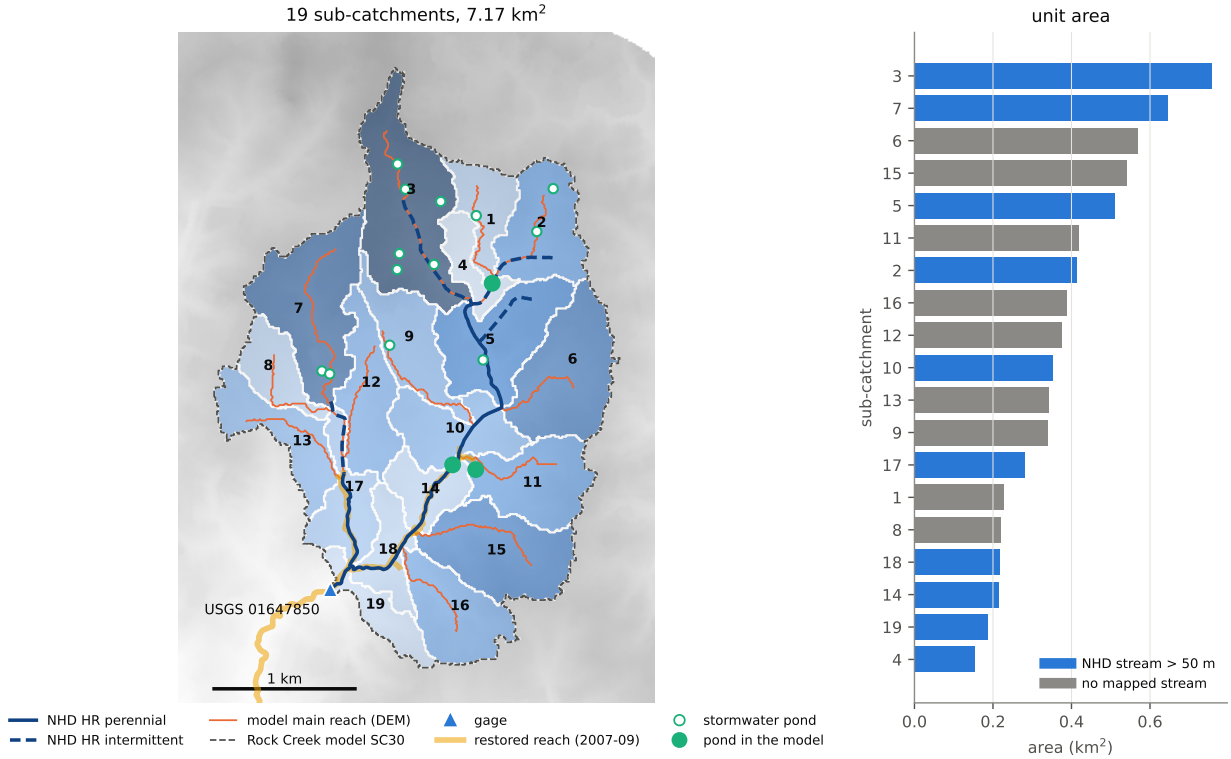


Figure 1: The 19 sub-catchments with their main flow paths, the mapped (NHDPlus HR) streams, the restored reach and the stormwater ponds; right: sub-catchment areas (grey: less than 50 m of mapped stream).

3 Model structure

3.1 Sub-catchments

The watershed is delineated from the USGS 3DEP 10 m elevation model, with the NHDPlus High Resolution flowlines burned in, into 19 sub-catchments whose outlets are the gage and the main confluences (Fig. 1). Each sub-catchment drains to one segment. Bed elevations come from a monotone profile carved through the elevation model, which passes road embankments and culverts (Fig. 2).

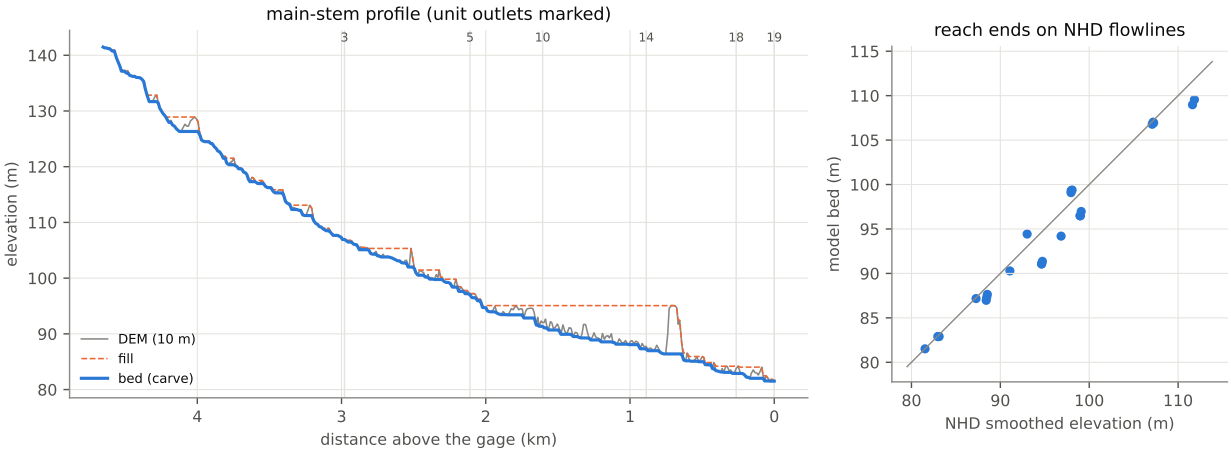


Figure 2: Main-stem longitudinal profile: elevation model, filled profile and model bed (carved).

3.2 The hydrologic response unit

Each sub-catchment is one *Urban HRU*, a composite element of OHQ (Fig. 3):

- an **impervious surface** — the *directly connected* impervious area (roofs, roads and parking areas that drain to a storm sewer or the stream) — that drains through an internal impervious reach to the unit's channel or street;
- a **pervious surface** — lawns, forest and the impervious surfaces that drain onto them — that drains through an internal pervious reach;
- three **soil layers** below the pervious surface, 0.5, 1 and 2 m thick, with their own hydraulic properties;
- an **unconfined aquifer** below the whole unit that receives recharge from the soil, discharges to the unit's channel or storm sewer (baseflow) and exchanges groundwater with the neighbouring units.

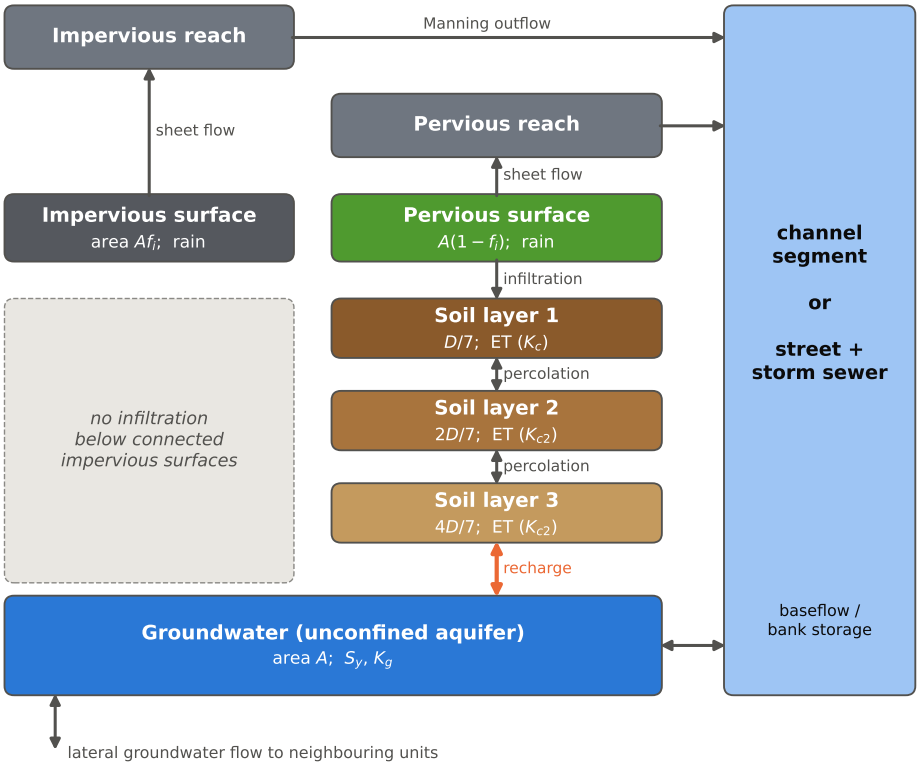


Figure 3: Structure of one sub-catchment (Urban HRU): storages (boxes) and fluxes (arrows; double arrows can flow both ways). A: area; f_i: connected impervious fraction; $D = 3.5\,\mathrm{m}$: soil depth.

3.3 Open channels and the gage section

Eight sub-catchments drain to open channels: the main stem from unit 4 down, the western tributary and one unmapped channel. Their cross-sections are measured on lidar transects every 20 m: a power-law main channel, $A = aY^m$, up to the bank height and a widening section above it (Fig. 5). The urban channels are two to three times wider than the rural Maryland Piedmont bank-full relations (McCandless and Everett, 2002) would give, which is why measured sections are used (Fig. 4). At the gage a short *gage segment* has a section fitted to the continuous record of simultaneous discharge and gage height under the current USGS rating (Fig. 6); its depth above the zero-flow control is directly comparable to the USGS stage.

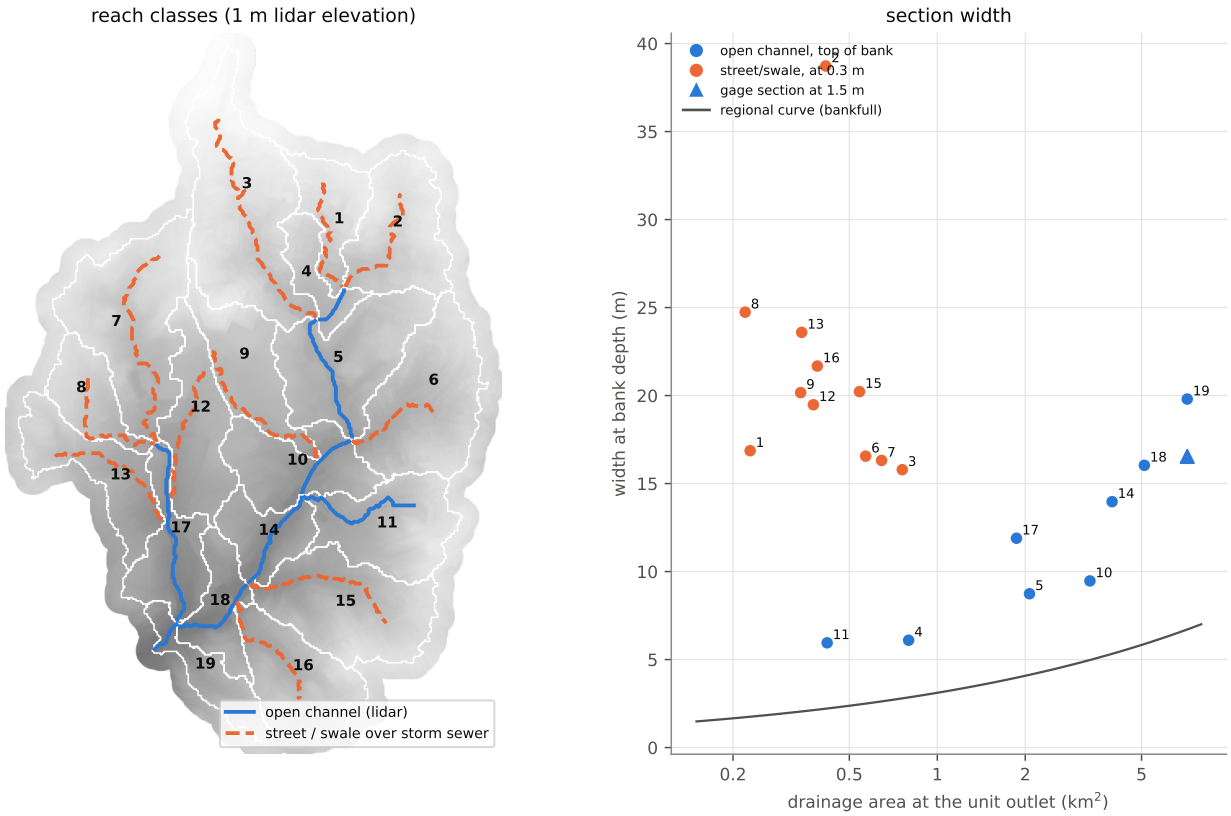


Figure 4: Left: open channels and piped units from the lidar transects. Right: channel width at bank height against drainage area, with the rural regional curve.

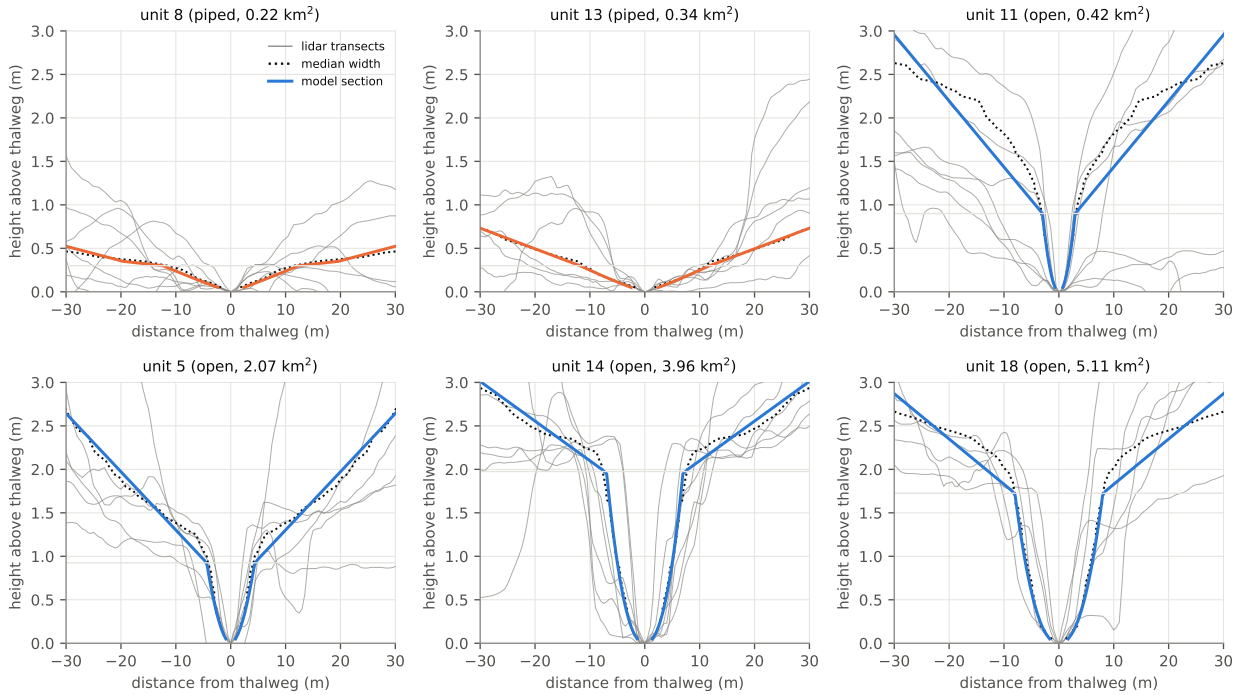


Figure 5: Lidar transects (grey), their median width (dotted) and the model section for two streets or swales and four open channels.

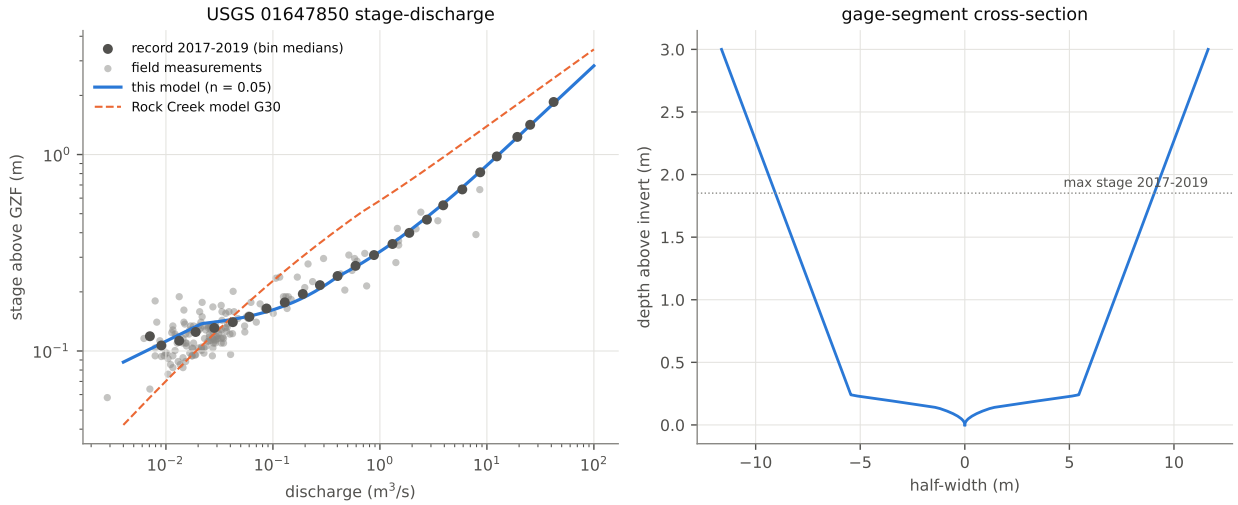


Figure 6: Stage against discharge at USGS 01647850 and the fitted gage section.

3.4 Streets and storm sewers

The other eleven sub-catchments have no channel: their surface flow path is a street or swale over a storm sewer. Each is represented by *dual drainage* (Fig. 9): runoff reaches the street segment, enters the sewer through the unit's inlets (lumped; weir or orifice flow, whichever is smaller), passes a catch basin that holds the unit's manholes and inlet boxes, and flows in the sewer to its outfall on the receiving channel or into the next sewer; what the inlets or the sewer cannot take continues

over the street. Where the sewer and the street are full, water spreads over the unit at the catch basin and drains back as the flood recedes. Sewer diameters are the recorded trunk sizes where the Montgomery County and State Highway Administration inventories give them, and otherwise follow a relation between pipe size and contributing area fitted to the 395 pipes of known size (Fig. 7); the fitted sizes are scaled by a calibrated multiplier. The groundwater of a piped unit discharges into its sewer.

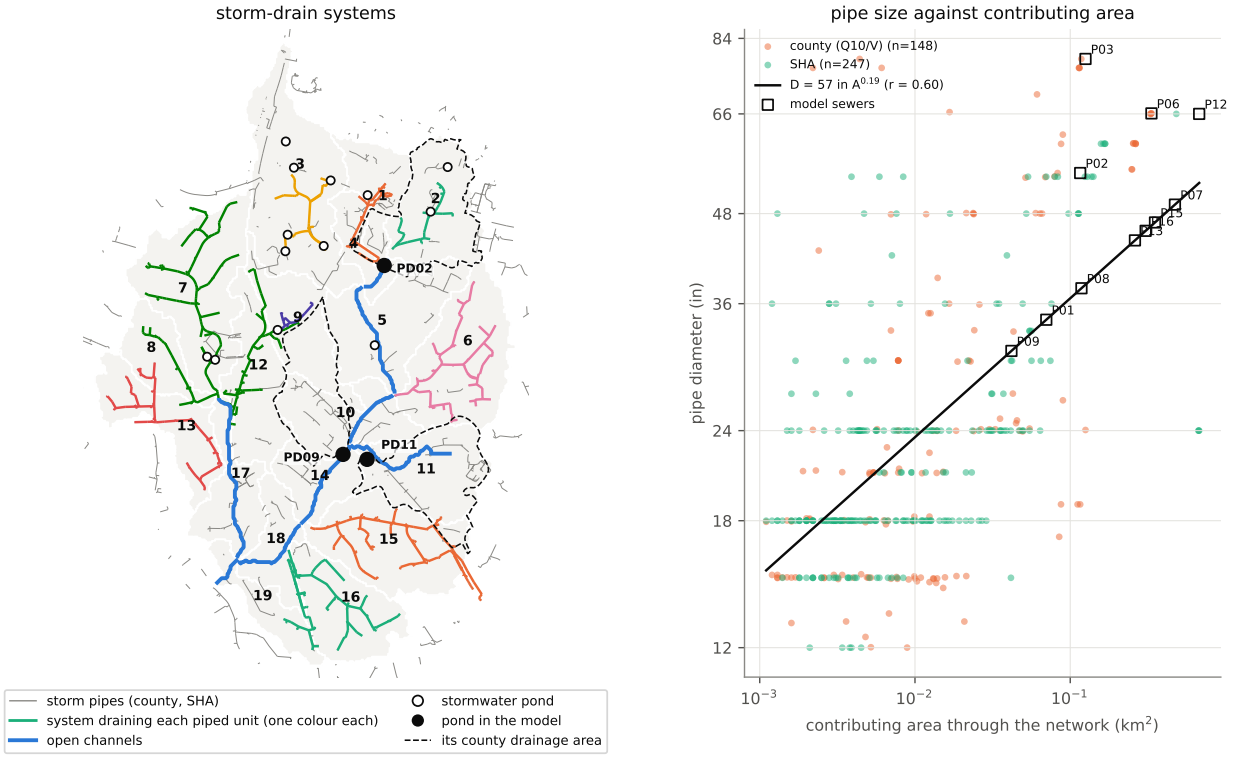


Figure 7: Left: storm pipes and the system that drains each piped unit; open channels in blue. Right: pipe diameter against contributing area, the fitted relation and the model sewers.

3.5 Stormwater ponds

Of the 16 stormwater ponds in the county's facility inventory, the three largest that each receive essentially one sub-catchment's outflow are represented: Peppertree (unit 2) and the two Matthew Henson ponds (units 9 and 11). Their stage–storage curves come from the 1 m lidar; their outlets (an extended-detention weir, a riser and an emergency spillway) follow Maryland design practice, since no as-built data were available (Fig. 8).

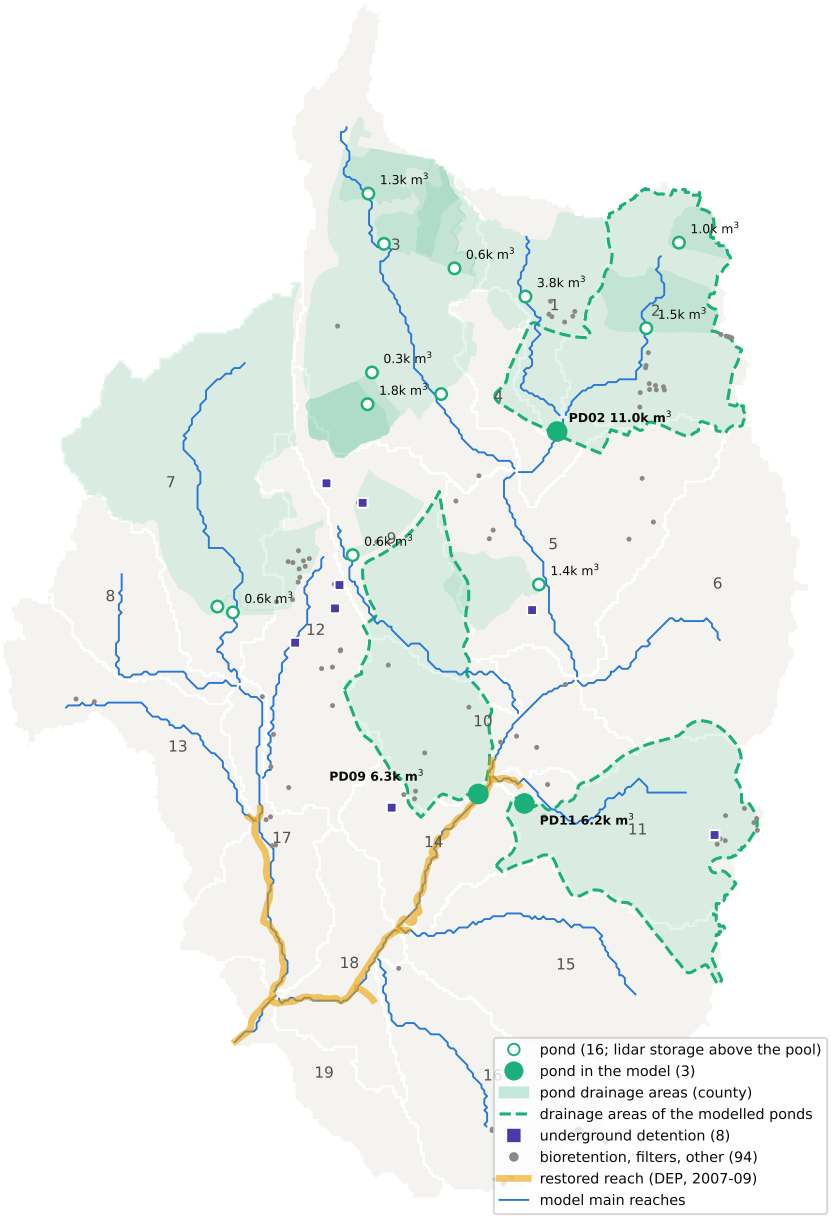


Figure 8: Stormwater facilities in the county inventory, the pond drainage areas (the three ponds in the model dashed and filled) and the restored reach.

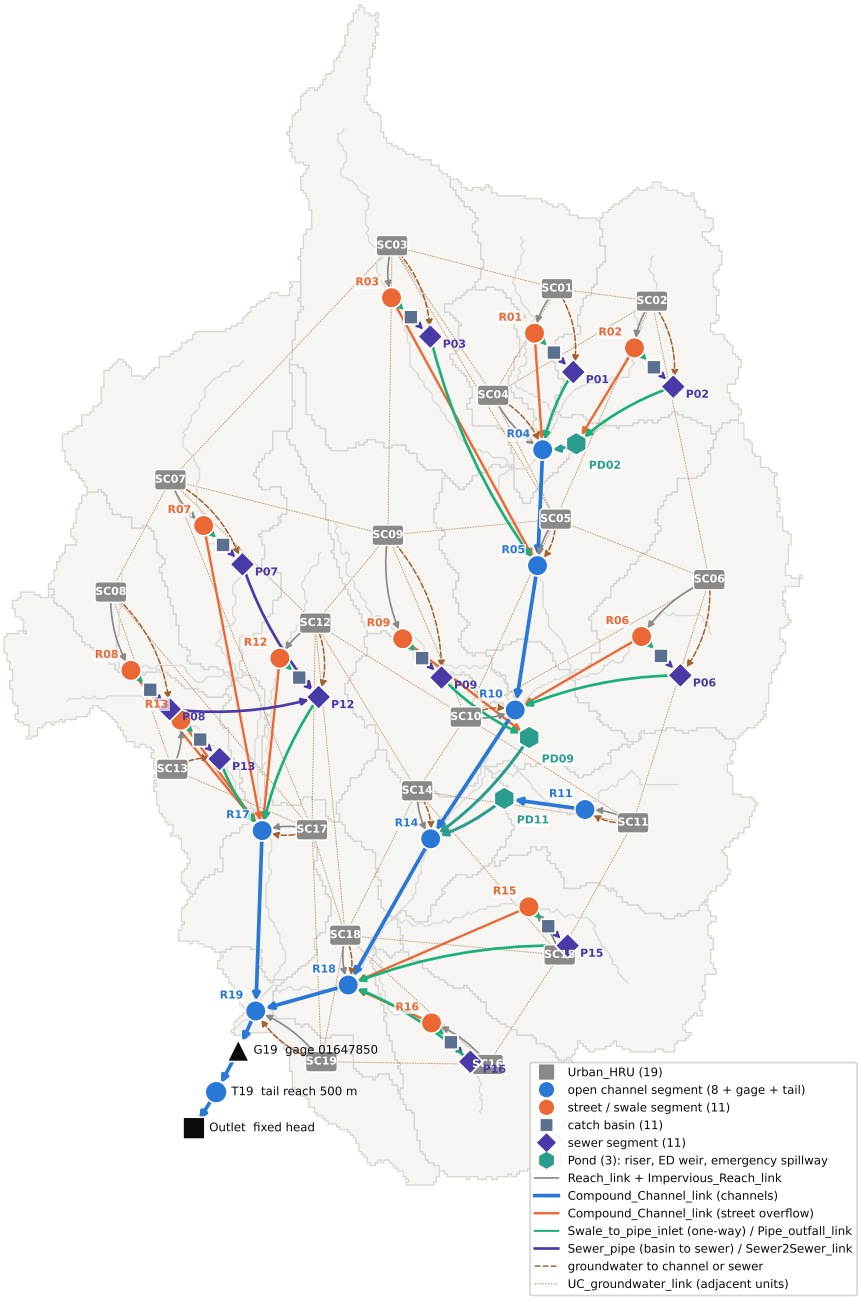


Figure 9: The whole model: sub-catchments (grey), open channels (blue), streets and swales (orange), sewers (purple), ponds (green), the gage segment, a tail reach and the outlet, with their surface, channel, inlet, outfall, groundwater and lateral groundwater connections.

4 Governing equations

OpenHydroQual represents a system as a network of storage elements (blocks) connected by flux elements (links), each defined by user-editable expressions, and solves the resulting equations to-

gether (Massoudieh et al., 2023; the equations of the standard components are given in Massoudieh and Shakouri, 2025). It solves a water balance for every storage element (surface, internal reach, soil layer, aquifer, channel, street, sewer, catch basin, pond),

$$\frac{dV_j}{dt} = \sum_{k \in \mathrm{in}(j)} Q_k - \sum_{k \in \mathrm{out}(j)} Q_k + P_j - E_j, \tag{1}$$

where V_j is the stored volume, Q_k the flows of the links that enter and leave it, P_j rainfall and E_j evapotranspiration. Lengths are in metres and time in days.

Surface runoff. Water ponded on a surface above its depression storage d_s runs off as sheet flow, $Q_s = (W/n_s)\, S^{1/2}\, (d - d_s)_+^{5/3}$, with overland-flow width W, slope S and Manning coefficient n_s.

Reaches, channels and streets. Flow between segments follows Manning's equation driven by the difference in water-surface elevation, $Q = \bar{n}^{-1} A_u R^{2/3} \big((h_1 - h_2)/\bar{L}\big)^{1/2}$, so backwater and flow reversal are represented.

Inlets, sewers and outfalls. Inlet flow is the smaller of the weir flow $C_w P\, d^{3/2}$ and the orifice flow $C_o A_o \sqrt{2g\,\Delta h}$ (HEC-22), with curb-opening length P and opening A_o of the unit's inlets. Sewers carry Manning flow in a circular section and surcharge when full; outfalls discharge with the sewer's flowing area and are free above the tailwater.

Soil water. Each soil layer follows the van Genuchten–Mualem relations,

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}, \quad \psi = -\frac{1}{\alpha}\left(S_e^{-1/m} - 1\right)^{1/n}, \quad k_r = \left[1 - \left(1 - S_e^{1/m}\right)^{m}\right]^2, \quad m = 1 - \tfrac{1}{n}, \tag{2}$$

with total head $h = z + \psi$. Infiltration into the top layer is Darcy flow over half its thickness and percolation between layers is $Q = A\, K_s\, k_r(S_e)\, (h_i - h_j)/\bar{d}$, upward when the lower layer has the higher head.

Evapotranspiration. Each soil layer loses $E = K_c\, E_0(t)\, A\, \theta/\theta_s$, where E_0 is the reference (grass) evapotranspiration and K_c a crop coefficient, one for the top layer and one for the two deeper layers.

Recharge and groundwater. The bottom soil layer drains to the aquifer through its unsaturated conductivity, and receives water from it when the water table is higher. The water table follows from storage and specific yield S_y, $h_g = z_b + V/(S_y A)$; groundwater moves between neighbouring units by Darcy's law, $Q = \bar{K}\, \bar{b}\, W\, (h_1 - h_2)/L$, with the shared boundary length W and centroid distance L, and lateral outflow tapers to zero as a cell approaches the depth of its stream bed.

Baseflow. The aquifer exchanges water with the channel or sewer, $Q_b = K_b\, A_b\, (h_g - h_c)/L_b$, with bank area A_b (twice the reach length times 5 m) and path length L_b (half the inverse drainage density).

Ponds. Storage $S = \alpha d^{\beta}$ from the lidar; the riser discharges the lesser of the weir and the barrel capacity, the emergency spillway is a broad-crested weir.

Numerical solution. The equations of all 199 storages are solved together with an implicit (backward Euler) scheme and Newton iterations, with an adaptive time step of 43 s to about 7 minutes. For speed the model is compiled into C++ by the OHQ code generator, which reproduces the interpreter's solution; four years take about one minute on one processor core.

5 Data

All inputs are public (Table 1).

Table 1: Data sources.

Model input	Source
Terrain, drainage network	USGS 3DEP 10 m DEM and 1 m lidar (MD Central 2024); NHDPlus High Resolution
Land cover, imperviousness	Chesapeake Conservancy 1 m land use/land cover
Connected impervious area	Sutherland (2000) relation
Soils	NRCS SSURGO; van Genuchten parameters from Rosetta3
Channels	1 m lidar transects; USGS continuous discharge and gage-height record
Storm sewers, inlets	Montgomery County and Maryland SHA storm-drain inventories
Stormwater ponds, restoration	Montgomery County DEP stormwater-facility inventory and project database; 1 m lidar
Rainfall (calibration, validation)	NOAA Analysis of Record for Calibration (AORC), hourly, 1 km
Rainfall (operational)	NOAA MRMS multi-sensor QPE (hourly, 1 km); Open-Meteo forecast
Reference ET	gridMET (daily, 4 km); Open-Meteo FAO-56 reference ET (recent days and forecast)
Discharge and stage	USGS National Water Information System, gage 01647850

Land-cover fractions of each sub-catchment come from the 1 m Chesapeake land cover; total impervious area is converted to directly connected impervious area with the "average" Sutherland relation $\mathrm{DCIA} = 0.1\,\mathrm{TIA}^{1.5}$ (in percent; 25 % for the basin), which the calibration then scales watershed-wide. Soil hydraulic properties of the three layers are area- and depth-weighted SSURGO values (the basin is 82 % hydrologic soil group B), with van Genuchten parameters from Rosetta3. Overland-flow length, width and slope come from the terrain (Fig. 10).

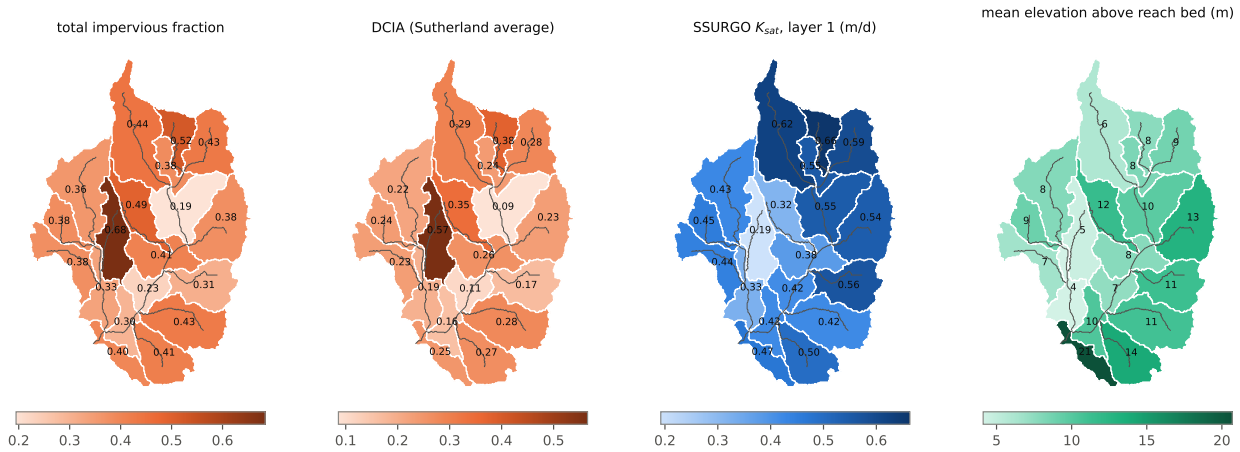


Figure 10: Total impervious fraction, connected impervious fraction, SSURGO saturated conductivity of the top soil layer and mean ground elevation above the stream bed, by sub-catchment.

Meteorological forcing. Rainfall and reference evapotranspiration are averaged over each sub-catchment with the overlap areas of the grid cells; mean annual AORC rainfall is about 1250 mm. The model was calibrated and validated with AORC rainfall; operationally it uses MRMS, multiplied by 1.08, the ratio of AORC to MRMS found over October 2024 to September 2025 for the surrounding Rock Creek watershed.

6 Calibration

Ten parameters were estimated, each applied identically to all 19 sub-catchments (Table 2), so that the differences between sub-catchments come entirely from the data above. They were fitted to discharge at the gage over water years 2022–2025 and to stage over 2024–2025 (the period of the current USGS rating), using hourly values during storms and daily values otherwise, after a two-month spin-up. The errors of the logarithms of flow and stage are modelled as autocorrelated (a first-order autoregressive process in continuous time) with standard deviations and correlation times that were estimated with the parameters.

The parameters were estimated by Bayesian inference: a genetic algorithm and the Levenberg–Marquardt method located the best fit, and an adaptive Markov chain Monte Carlo sampler (23 chains) then drew samples from the posterior distribution, starting from it. The sampler found a better region than the gradient method (the log-likelihood improved by about 530), mainly by dropping the large extra factor on the conductivity of the top soil layer (from 26 to 1.1). The sampler ran 1722 steps per chain; the posterior is taken from the last 722 (16 606 samples), after the chains had mixed (Gelman–Rubin $\hat{R}$ 1.04–1.23, effective sample size 260–450 per parameter). The numerical solver's time-step growth was limited to a factor of 2 per step in the final stages and in the twin: with larger growth, tiny parameter changes altered the sequence of time steps and made the likelihood too rough for the sampler, without changing the fit. The twin runs the sample with the highest posterior probability; Table 2 gives it with the 95 % posterior range.

Table 2: Calibrated parameters: the twin's parameter set and the 2.5–97.5 % posterior range.

Parameter	Role	Twin	95 % range
Soil conductivity multiplier	scales SSURGO K_s of all layers	0.032	0.030–0.038
Surface conductivity multiplier	additional factor on the top layer	1.11	0.85–1.26
Connected-impervious multiplier	scales the Sutherland DCIA	0.39	0.38–0.41
Crop coefficient, top layer	K_c	0.53	0.50–0.58
Crop coefficient, deeper layers	K_{c2}	0.13	0.11–0.14
Aquifer–stream conductivity	K_b, m/d	254	207–327
Aquifer conductivity	K_g, m/d (lateral flow)	550	444–857
Aquifer specific yield	S_y	0.0066	0.0056–0.0072
Channel roughness	Manning n, open channels	0.084	0.075–0.093
Sewer diameter multiplier	on the fitted sewer sizes	0.44	0.39–0.47
Flow error	s.d. of log flow; correlation time (d)	0.54; 0.12	
Stage error	s.d. of log stage; correlation time (d)	0.17; 0.13	

The soil conductivity multiplier is small: the calibration reduces the SSURGO conductivities about 30-fold, which lets the pervious surfaces produce runoff in large storms (SSURGO conductivities exceed most hourly rain rates). The connected-impervious multiplier means that about 10 % of the basin, rather than the 25 % from the Sutherland relation, sends its runoff directly to the streams and sewers, plausibly reflecting the stormwater facilities and disconnected roofs that the model does not represent individually. Some parameters reach or pass their prior bounds (the sewer multiplier lies below its lower bound of 0.5; the specific yield and the aquifer conductivity are close to theirs); the priors are deliberately not truncated.

6.1 How the uncertainty is quantified

Error model and likelihood. A measured flow or stage y_i at time t_i is related to the model value $f_i(\boldsymbol{\theta})$ for parameters $\boldsymbol{\theta}$ by

$$\ln y_i = \ln f_i(\boldsymbol{\theta}) + \varepsilon_i, \qquad \varepsilon_i = \phi_i\,\varepsilon_{i-1} + \sqrt{1-\phi_i^2}\;\sigma\,\eta_i, \qquad \phi_i = e^{-(t_i - t_{i-1})/\tau}, \quad \eta_i \sim N(0,1), \tag{3}$$

a multiplicative (log-normal) error that is autocorrelated in time: a continuous-time first-order autoregressive process with standard deviation σ and correlation time τ, separately for flow and stage. Errors one hour apart in a storm are strongly correlated ($\phi \approx 0.7$), errors a day apart nearly independent. The likelihood is the product of the normal densities of the whitened residuals $(\varepsilon_i - \phi_i\varepsilon_{i-1})/(\sigma\sqrt{1-\phi_i^2})$ over the 9144 flow and 4808 stage measurements, with the Jacobian of the transformation; σ and τ are estimated with the model parameters. Treating the errors as independent would count each storm's hourly points as independent evidence and make the posterior far too narrow.

Prior. Each parameter has a log-normal prior (normal for the two crop coefficients) whose ± 2 standard deviations span the range in Table 3: wide, physically plausible ranges, from four to seven orders of magnitude for the conductivities. The priors are smooth and not truncated: the data may move a parameter beyond its range if the evidence is strong, which happened for the sewer diameter multiplier.

Sampling. The posterior $p(\boldsymbol{\theta} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\theta})\,p(\boldsymbol{\theta})$ was sampled with OpenHydroQual's adaptive Metropolis sampler: 23 chains, each proposing a jump from its current state drawn from a multi-

variate normal shaped by the covariance of the chain's own past samples and scaled to keep about a fifth of the proposals accepted. Each proposal requires one run of the compiled model over the four calibration years (about 3 min), so the 39 606 samples took about ten days on 23 processor cores. The chains were started at the best fit of the gradient method, moved together to a region of higher probability within about 300 steps, and were run for 1722 steps (Fig. 11). The statistics use the last 722 steps, over which nothing drifts. Two diagnostics confirm that the chains sample the same distribution: the Gelman–Rubin statistic $\hat{R}$, which compares the spread between chains with the spread within them (1 when they agree; here 1.04–1.23), and the effective sample size, the number of independent draws the autocorrelated samples are worth (260–450 per parameter, enough for 95 % intervals).

A numerical detail. The model's solver lengthens its time step after each successful step. With a growth of up to a factor of 10 per step, tiny parameter changes altered the sequence of time steps during storm rises and moved the log-likelihood by several units, which stalled the sampler; with growth limited to a factor of 2 the likelihood is smooth and the fit unchanged. The final sampling stages and the twin use the factor of 2.

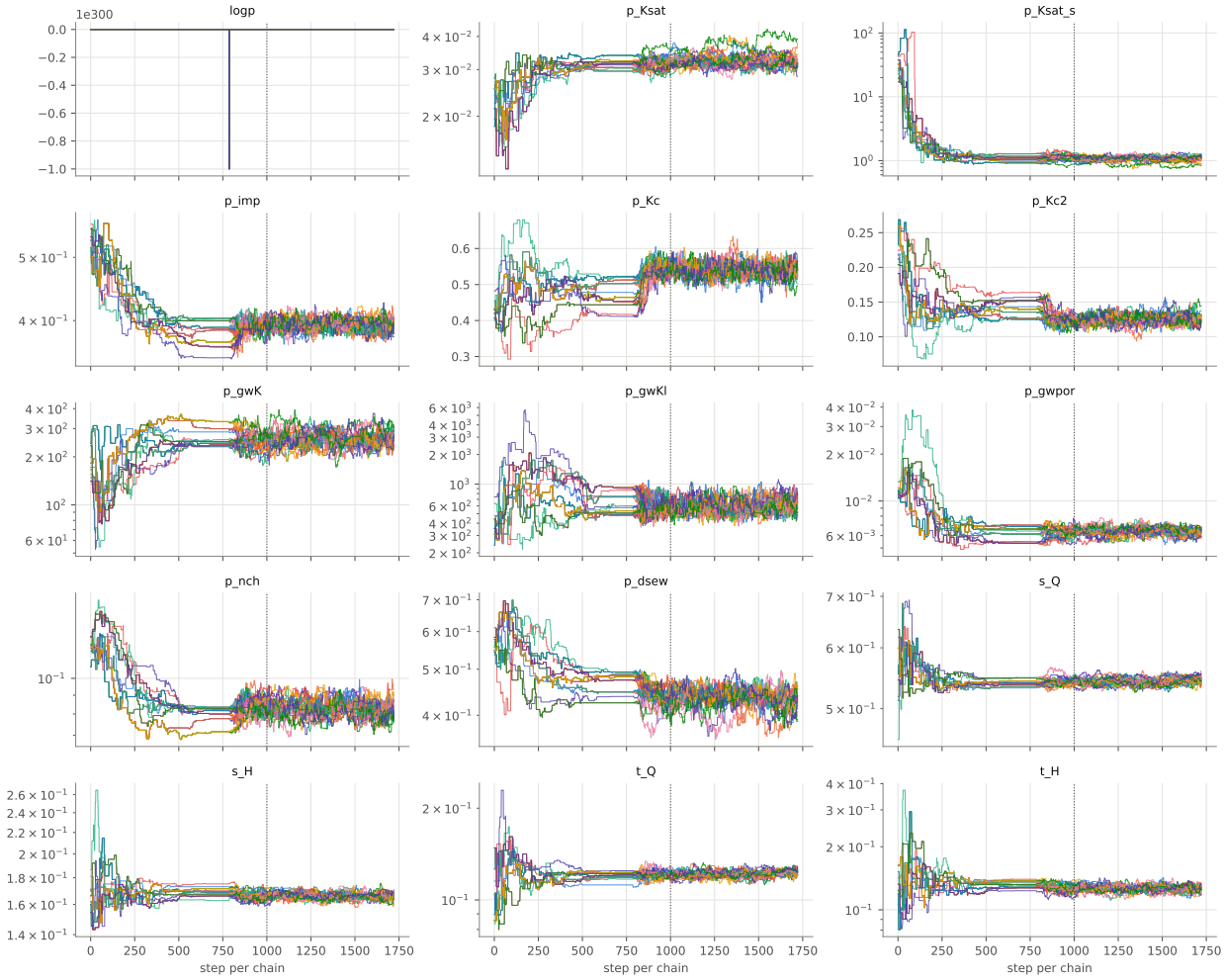


Figure 11: Traces of the 23 chains: log-posterior (top left) and each parameter against the step of the chain. The dotted line marks the start of the samples used (step 1000); the spike at step 787 is the restart with the smoother time stepping, whose first proposal was accepted unconditionally.

6.2 Posterior distribution

The sampler used an adaptive Metropolis method (proposals shaped by the covariance of each chain's samples) with the autocorrelated log-normal likelihood above and log-normal priors (normal for the crop coefficients) whose ±2 standard deviations are the parameter ranges; the priors are smooth and not truncated at the ranges. The 23 chains started at the best fit of the gradient method, moved together to the region of higher probability within about 300 steps and were then run on; the statistics use the last 722 steps of each chain (16 606 samples), over which the chains agree (Gelman–Rubin $\hat{R}$ 1.04–1.23; the largest values for the two soil conductivity multipliers, which can partly compensate each other) and the samples are equivalent to 260–450 independent draws per parameter. The posterior is narrow (Table 3 and Fig. 12): the 95 % intervals extend from 2 % either side of the median (error standard deviations) to 5–25 % (soil, impervious, evapotranspiration, aquifer and channel parameters), and the twin's parameter set lies inside every interval.

Table 3: Posterior distribution of the parameters: the twin's set (highest posterior probability), median and 95 % interval, and the prior range (±2 standard deviations).

Parameter	Twin	Median	95 % interval	Prior range
Soil conductivity multiplier	0.032	0.032	0.030–0.038	0.001–10
Surface conductivity multiplier	1.11	1.09	0.85–1.26	0.01–1000
Connected-impervious multiplier	0.39	0.39	0.38–0.41	0.3–3
Crop coefficient, top layer	0.53	0.54	0.50–0.58	0.3–1.4
Crop coefficient, deeper layers	0.13	0.13	0.11–0.14	0–1
Aquifer–stream conductivity (m/d)	254	256	207–327	0.1–10 000
Aquifer conductivity (m/d)	550	571	444–857	0.01–1000
Aquifer specific yield	0.0066	0.0064	0.0056–0.0072	0.005–0.4
Channel roughness (Manning n)	0.084	0.083	0.075–0.093	0.02–0.2
Sewer diameter multiplier	0.44	0.44	0.39–0.47	0.5–4
Error s.d., log flow	0.54	0.54	0.53–0.55	0.05–3
Error s.d., log stage	0.16	0.17	0.16–0.17	0.02–2
Error correlation time, flow (h)	2.8	2.9	2.8–3.1	0.024–720
Error correlation time, stage (h)	3.0	3.0	2.8–3.3	0.024–720

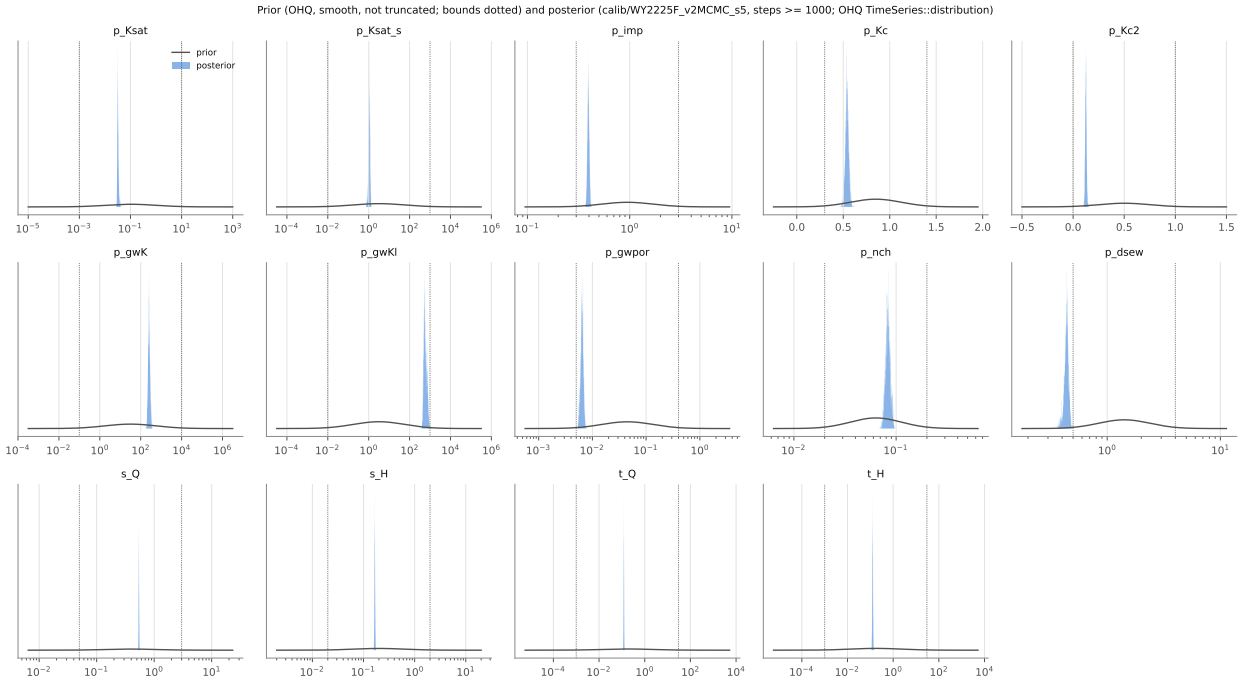


Figure 12: Prior (line; bounds dotted) and posterior (histogram) of each parameter. Parameters with a log-normal prior on a logarithmic axis. Every posterior is much narrower than its prior.

The parameters are not independent (Fig. 13). Three pairs are strongly correlated because they act on the same part of the hydrograph and can partly replace each other: the soil and surface conductivity multipliers (−0.93; both set the infiltration capacity of the top soil), the two crop coefficients (−0.93; together they set the total evapotranspiration), and the aquifer–stream conductivity and the channel roughness (−0.81; both shape the recession). These trade-offs are why the individual intervals are wider than the constraint on the flow itself; the twin keeps them by running whole parameter sets drawn from the posterior, never parameters varied one at a time.

The error standard deviation and correlation time of each series are also correlated (0.7).

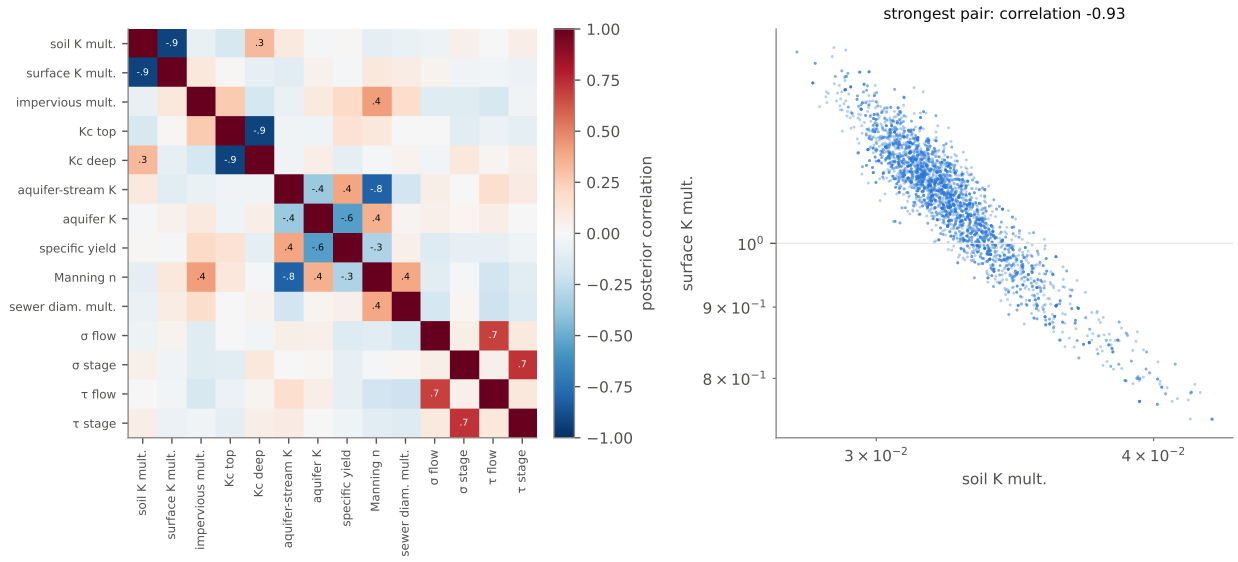


Figure 13: Left: posterior correlation of the parameters (logarithms for parameters with a log-normal prior; values of 0.3 or more in magnitude printed). Right: the strongest pair, soil against surface conductivity multiplier, 4000 posterior samples.

7 Fit and validation

The twin's parameter set was run continuously over the calibration period (WY2022–2025) and, unchanged, over WY2018–2021, which was not used in calibration (Tables 4 and 5 and Figs. 14 and 15). The logarithm of flow, which weights low flows and recessions, is reproduced with an NSE of 0.83–0.84 in both periods and stage with 0.79–0.81. The NSE of the flows themselves, dominated by the largest storms, is 0.65–0.68: storm peaks are about 25 % high on average and individual convective storms are missed or over-predicted, which is also the effect of the 1 km rainfall grid on a basin of this size. Low flows are 1.4–1.6 times the observed, the same tendency as in the Rock Creek model. The fit varies little between water years (log NSE 0.82–0.85).

Table 4: Fit of the twin's parameter set at USGS 01647850. The measures are computed on the observation points (hourly in storms, daily otherwise).

Period		NSE	logNSE	KGE	Volume	Low 20 %	High 20 %	n
Calibration WY2022–25	flow	0.65	0.84	0.65	1.20	1.57	1.25	9144
	stage	0.76	0.81	0.80				4808
Validation WY2018–21	flow	0.68	0.83	0.69	1.22	1.43	1.25	9558
	stage	0.69	0.79	0.74				4549

Table 5: Flow fit by water year (val.: validation, cal.: calibration).

Water year	2022	2023	2024	2025	2018	2019	2020	2021
Flow NSE	0.67	0.51	0.61	0.72	0.58	0.55	0.79	0.64
Flow logNSE	0.85	0.84	0.84	0.82	0.83	0.83	0.82	0.83
Period	cal.	cal.	cal.	cal.	val.	val.	val.	val.

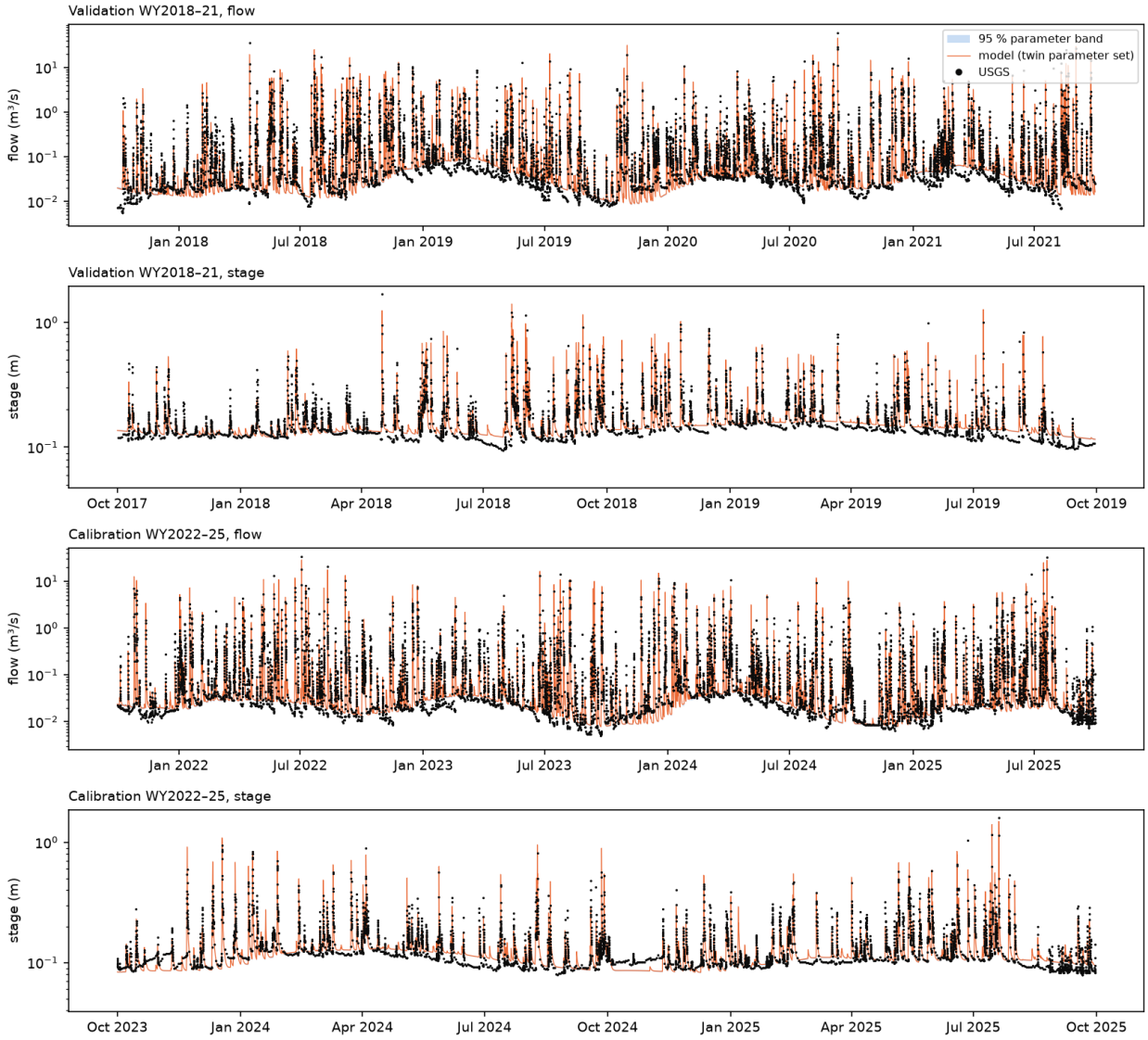


Figure 14: Validation (WY2018–21) and calibration (WY2022–25) periods: USGS (black), the twin's parameter set (orange) and the 95 % parameter band (blue), log scale.

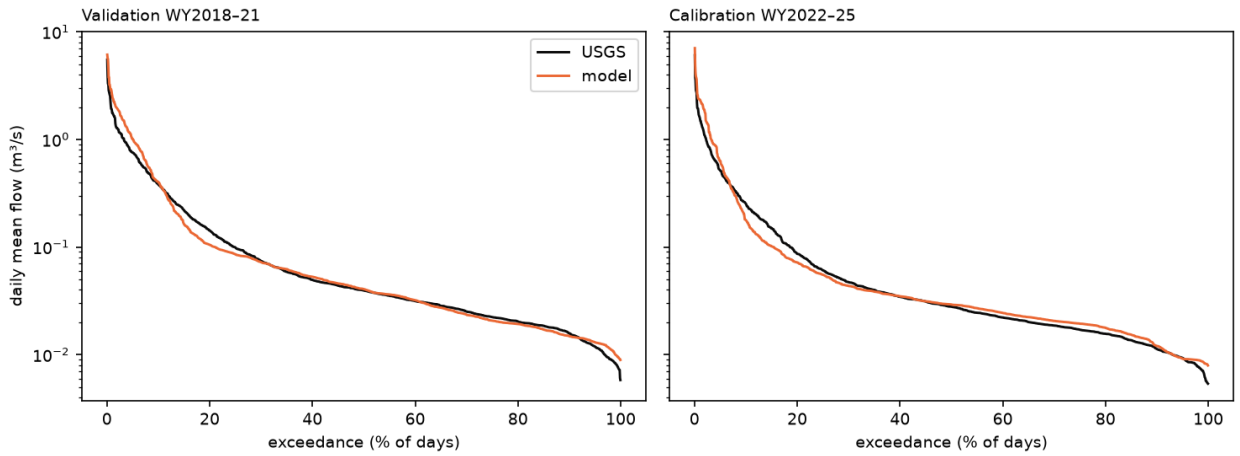


Figure 15: Flow-duration curves of the daily means of the observation points: USGS and the twin's parameter set.

Fit measures. NSE: Nash–Sutcliffe efficiency (1 is perfect, 0 is no better than the mean); logNSE: the same on the logarithms; KGE: Kling–Gupta efficiency, which combines correlation, variability and bias; volume: total model over observed; low/high 20 %: model over observed mean of the lowest/highest 20 % of the observed values.

7.1 Uncertainty

One hundred parameter sets drawn from the posterior were run over both periods (Table 6 and Figs. 16, 17 and 19). The median of these runs fits as well in the validation years as in the calibration years (log-flow NSE 0.83 and 0.84), so the model is not tuned to the calibration period. The 95 % parameter interval, the darker band of the twin, is narrow: its median width is a factor of 1.09 on flow and 1.02 on stage, and it contains only 6–8 % of the measurements. Nearly all of the difference between model and measurements is therefore not parameter uncertainty but error of the model structure and the rainfall. Adding the calibrated error model (log-normal with autocorrelation) gives the 95 % prediction interval, which contains 92–93 % of the measurements in both periods, close to the nominal 95 %: more than 98 % in dry weather, where it is wider than needed, and 91–92 % during storms, where it is slightly too narrow. The twin shows both intervals; the uncertainty of the rainfall forecast, the largest source of forecast error, is not yet represented.

Table 6: 95 % intervals from 100 posterior parameter sets: log NSE of the median run, share of the measurements inside each interval (prediction interval: all points, storm hours, daily points) and median width of each interval (upper/lower).

Period		log NSE	Inside the interval				Width (factor)	
			parameter	prediction	storms	daily	parameter	prediction
Calibration WY2022–25	flow	0.84	8 %	92 %	91 %	99 %	1.09	7.7
Calibration WY2024–25	stage	0.81	6 %	93 %	92 %	99 %	1.02	1.9
Validation WY2018–21	flow	0.83	7 %	92 %	92 %	98 %	1.09	7.6
Validation WY2018–19	stage	0.79	6 %	93 %	92 %	100 %	1.02	1.9

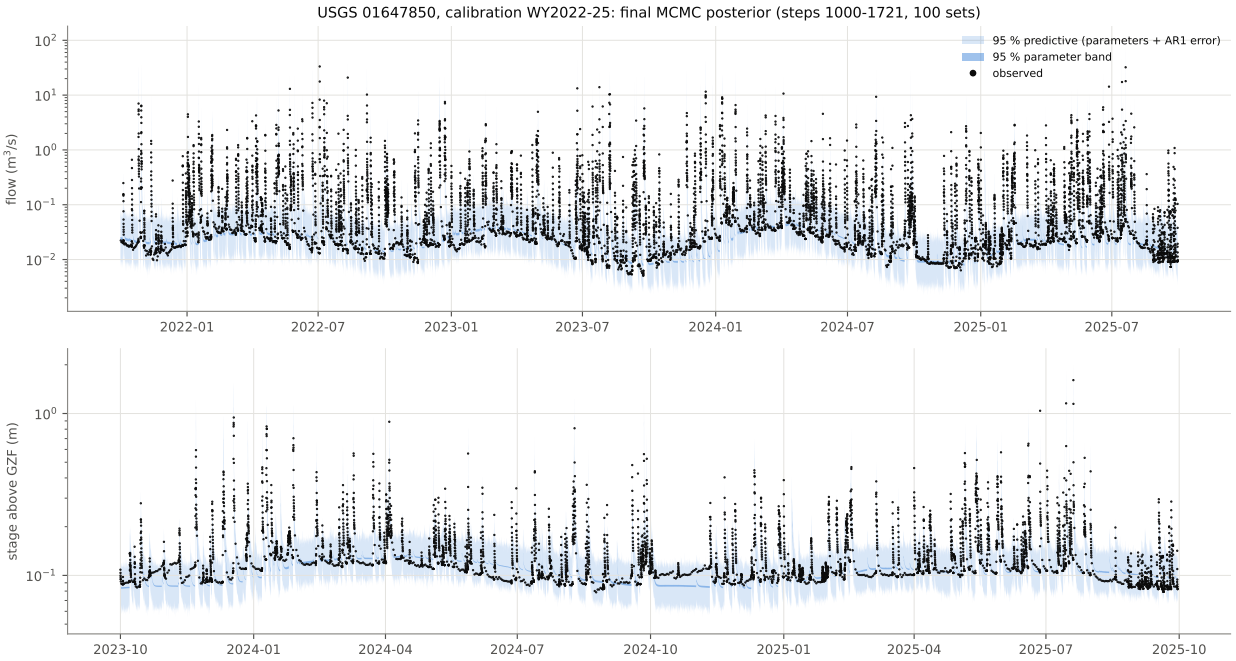


Figure 16: Calibration years: measured flow and stage (black) with the 95 % parameter interval (darker) and the 95 % prediction interval (lighter) from 100 posterior parameter sets.

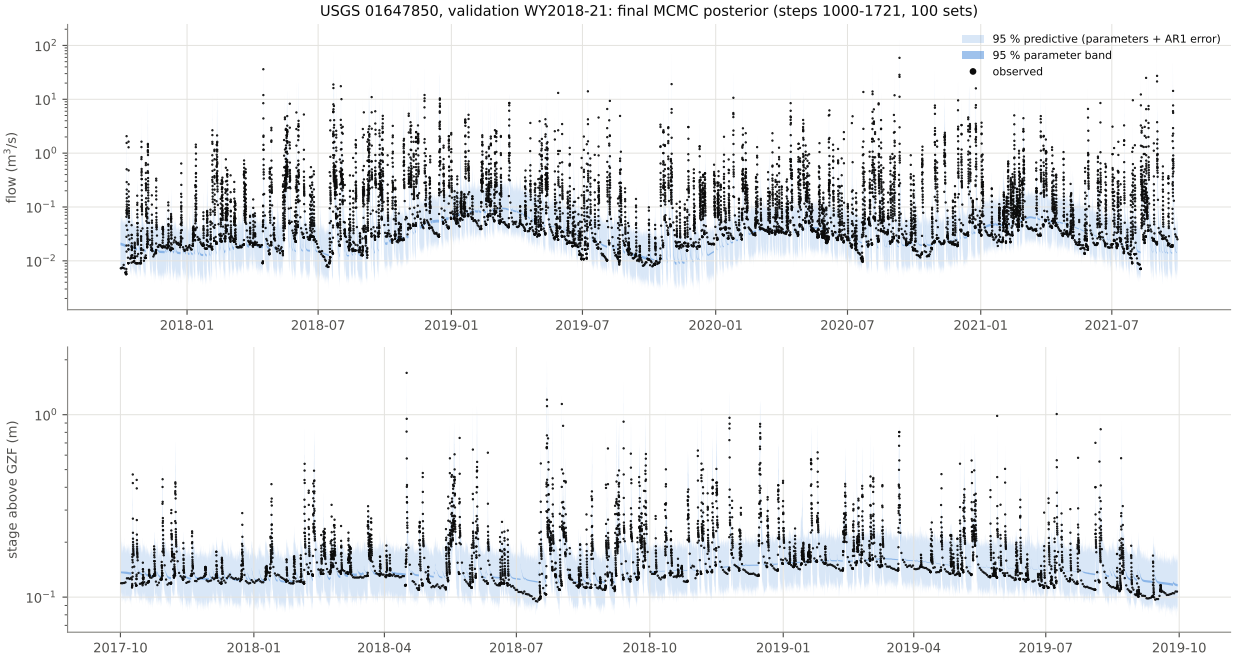


Figure 17: As Fig. 16, validation years (not used in the calibration).

Is the prediction interval reliable? An interval is reliable when the share of measurements inside it equals its nominal level, at every level. Figure 18 checks the interval as the twin draws it (the median of the posterior runs times $\exp(\pm z\sigma)$, $\sigma^2 = \sigma_e^2 + \sigma_p^2$): the curves of the observed against

the nominal coverage lie close to the diagonal for flow and stage, in calibration and validation alike; the 95 % interval contains 93–94 % of the measurements and the 50 % interval 50–55 %. (Simulating the autocorrelated error explicitly, as in Table 6, gives 92–93 %.) The interval is reliable across flow magnitudes except at the top: in the highest tenth of the measured flows it contains 84 % of the measurements, because the error of large storms, driven by the rainfall on a basin covered by a few radar cells, is larger than one standard deviation for all flows can represent. The standardized residuals follow the normal shape assumed by the likelihood, with somewhat heavier tails; stage residuals are centred slightly below zero (the model's low-flow stage is a little high).

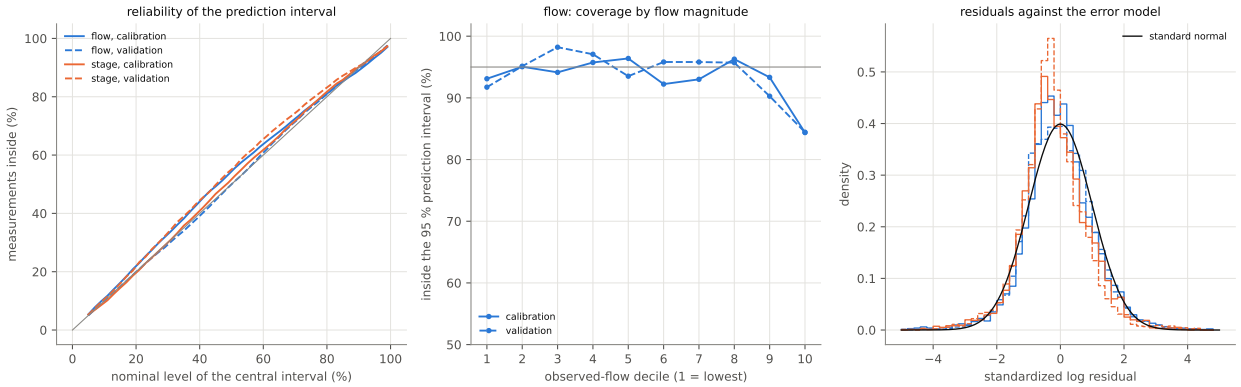


Figure 18: Reliability of the prediction interval. Left: share of measurements inside the central interval against its nominal level (diagonal: perfect). Middle: share of flow measurements inside the 95 % interval by decile of the measured flow. Right: standardized log residuals $(\ln y - \ln f)/\sigma$ against the standard normal density. Solid: calibration, dashed: validation; blue: flow, orange: stage.

How the twin carries the uncertainty forward. Each of the 200 parameter sets keeps its own model state (soil moisture, groundwater, channel and pond storage) from cycle to cycle, so the parameter interval of the forecast includes the different wetness each parameter set has built up, not only its different response to the next storm. The prediction interval is added afterwards, for streamflow and gage stage, with the calibrated error standard deviations. Neither interval includes the uncertainty of the rainfall forecast; for that, the rainfall forecast itself would have to be run as an ensemble.

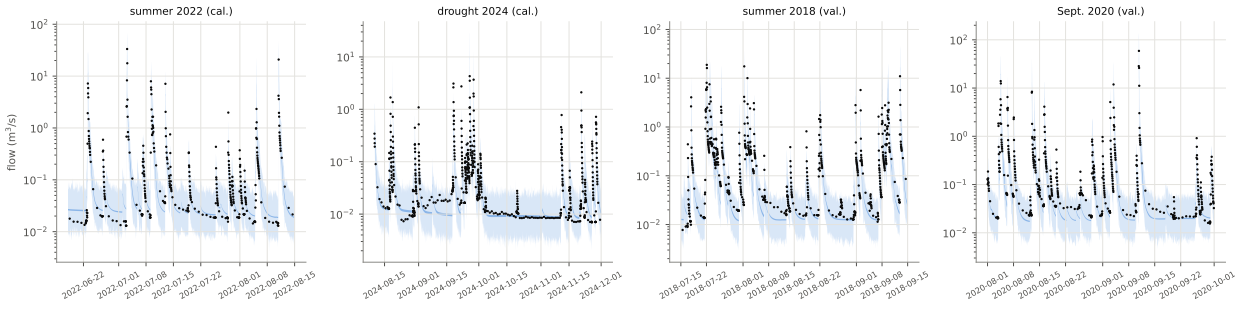


Figure 19: Flow in four periods of the calibration (cal.) and validation (val.) years, with both intervals.

8 Limitations

- **Low flows.** In dry weather the model keeps flow about 1.4–1.6 times the observed; low-flow stage is slightly high.
- **Storm peaks.** Peaks are on average about 25 % high, and the timing and size of individual convective storms depend on the radar rainfall over a basin covered by only a few 1 km cells.
- **Forecast rainfall** is the largest source of forecast error; the twin runs a single rainfall forecast and neither interval includes its uncertainty.
- **Simplifications.** Each piped unit's storm sewers are lumped into one sewer; small piped transfers across sub-catchment divides are neglected; only the three largest ponds are represented, with assumed outlets; bioretention and other small facilities are not represented individually.
- **No data assimilation** yet: the model state is not corrected with the real-time observations.

9 Software and reproducibility

The model is built and run with open-source software: OpenHydroQual (`https://github.com/ArashMassoudieh/OpenHydroQual`; Massoudieh et al., 2023; Massoudieh and Shakouri, 2025), the digital-twin engine OpenHydroTwin (`https://github.com/ArashMassoudieh/OHTwin`) and the watershed twin and viewer OpenWatershedTwin (`https://github.com/ArashMassoudieh/OpenWatershedTwin`). Every input file of the model is produced by numbered scripts from the public data listed in Table 1. The live twin is at `https://openhydrotwin.com/TurkeyBranch/`.

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